
From Signal Fusion to Asset Allocation: A Decision-Theoretic Framework for Regime-Conditional Portfolio Construction

Abstract: Predictive modelling in finance has advanced faster than the machinery that converts a prediction into a position. This paper specifies and tests a decision-theoretic framework for that conversion. Latent market states are identified by a Gaussian mixture model fitted on an expanding window; technical and market-state composites are fused with state-dependent weights and scaled to a return metric; the fused signal enters a constrained quadratic program with covariance shrinkage and an explicit turnover penalty; and the resulting portfolio is scaled by a regime-dependent risk budget. Evaluated on ten liquid multi-asset instruments over eighteen years under a walk-forward protocol with transaction costs, the framework attains a Sharpe ratio of 0.88 against 0.63 for an equal-weighted benchmark and reduces maximum drawdown from -36.1 to -19.0 per cent, at the cost of a lower compound return. It outperforms in all four stress episodes in the sample and in 18 of 18 specifications examined, though the Sharpe improvement is not statistically significant. The final 665 sessions postdate every specification decision and are reported as a holdout; on them the framework records a higher Sharpe ratio and a smaller maximum drawdown than all three benchmarks considered. Ablation at matched average exposure isolates the source: timing risk by market state improves both the Sharpe ratio and the drawdown with no change in mean return, whereas the signal fusion layer contributes nothing and the state-dependent weighting hypothesis is unsupported. The paper also identifies two failure modes in frameworks of this kind: an absolute volatility target that never binds, and a signal-to-allocation conversion in which mismatched units rather than information determine the solution.

Keywords: Portfolio Optimisation; Regime Detection; Risk Budgeting; Decision-Theoretic Framework; Signal Fusion; Ablation Analysis; Asset Allocation

Reference

Biographical notes:

1 Introduction

Quantitative finance has invested heavily in prediction and comparatively little in the step that follows it. A large body of work now forecasts returns, directions or volatilities with machine learning, and reports its success in statistical terms: accuracy, error, or the significance of a coefficient (Gu et al., 2020; Zhang et al., 2022). Far less work specifies what an investor should then hold. That second step is not a formality. A forecast becomes an investment only after it has been reconciled with a covariance structure, a risk budget, position limits and the cost of trading, and the translation can destroy the value of a forecast as easily as preserve it. This paper refers to the missing step as the *signal–allocation gap*.

1.1 *The Signal–Allocation Gap*

The gap is visible in how results are reported. Systematic reviews of fusion-based prediction find that statistical accuracy remains the dominant success criterion, with economic value reported inconsistently or not at all (Zhang et al., 2022). Where allocation is addressed, it is often through heuristics rather than optimisation: the IntPort framework of Sami et al. (2025), for instance, converts forecasts into weights by statistical averaging, without a covariance structure or an explicit risk objective. Reinforcement learning approaches optimise a policy directly and so avoid the translation problem, but the majority are formulated for a single instrument (Saijai and Pankwaen, 2026; Jiang et al., 2025), which leaves the portfolio-level question open.

This matters because the properties an investor cares about are portfolio properties. Diversification, drawdown and the stability of holdings are not visible at the level of an individual forecast, and DeMiguel et al. (2009) showed that optimisation which ignores estimation error can underperform naive equal weighting out of sample. Any framework that claims to bridge prediction and allocation must therefore be judged on portfolio outcomes under realistic frictions, not on the quality of its forecasts.

1.2 *Regime Dependence*

Financial markets do not behave homogeneously through time. Since Hamilton (1989), regime-switching models have been used to represent discrete shifts in the statistical behaviour of financial series, and Ang and Bekaert (2002) demonstrated that accounting for such shifts materially changes optimal international allocations. Kritzman et al. (2012) made the practical case that regime awareness alters what a dynamic strategy should do, not merely how it should be described. More recent work continues in this direction, combining regime identification with allocation rules (Li et al., 2025; Luo and Mulvey, 2026).

Two consequences follow for a framework of the kind proposed here. The first is that a predictive signal’s reliability is conditional: an indicator that is informative in a quiet trending market need not be informative when the market is disturbed. The second is that the appropriate amount of risk to carry is itself state-dependent, which is the premise behind volatility-managed strategies (Moreira and Muir, 2017; Hood and Raughtigan, 2025) and behind risk budgeting more generally (Asimit et al., 2025). A framework that detects regimes but acts on them in only one of these two ways will leave part of its own premise untested.

1.3 *What This Paper Does*

This paper specifies a three-stage decision-theoretic framework and tests it end to end. Latent market states are identified by a Gaussian mixture model fitted on an expanding window. Two composite signals—one technical, one derived from price and volume dynamics—are fused with regime-conditional weights and converted to expected excess returns using information-ratio scaling. The fused signal then enters a constrained quadratic program with Ledoit–Wolf covariance shrinkage (Ledoit and Wolf, 2004) and an explicit turnover penalty, and the resulting sleeve is scaled by a regime-dependent risk budget.

The framework is evaluated on ten liquid exchange-traded funds spanning equities, fixed income, credit, gold and real estate over eighteen years, under a walk-forward protocol with transaction costs. The evaluation is designed to answer not only whether the framework works but which of its parts is responsible, and it reports the answer in both directions. The regime risk budget is shown to be the effective component: at identical average exposure,

timing the de-risking by regime improves both the Sharpe ratio and the maximum drawdown. The signal fusion layer is shown not to be: the information coefficients of both composites are statistically indistinguishable from zero or significantly negative, and the regime-conditional weighting hypothesis is rejected by the data that motivated it.

Reporting the second finding is a deliberate choice. Component-level nulls are rarely published, which is one reason architectures accumulate layers whose contribution has never been isolated. Harvey and Liu (2015) and Valls and Steurer (2026) document how easily an unreported specification search inflates apparent performance in this setting, and DeMiguel et al. (2009) showed the practical cost of trusting an unvalidated component. An ablation that locates the source of a result is more useful to a practitioner than an aggregate number that does not.

1.4 Contributions

- A fully specified regime-conditional allocation framework, given in enough detail to be reproduced from the text alone: exact feature definitions, the mixture model and its label-anchoring requirement, the optimisation problem with all constraints, and the walk-forward protocol.
- A two-step separation of the allocation decision into relative composition and total risk. The paper shows that expressing the regime risk budget as a fraction of the portfolio's own volatility, rather than as an absolute volatility target, is what keeps the regime layer active: absolute targets calibrated for concentrated equity portfolios bind in one rebalance out of 224 on a diversified universe and render the layer inert.
- An ablation design that isolates component contributions at matched risk. Because scaling a portfolio toward cash changes both its return and its volatility, comparing a de-risked strategy with a fully invested benchmark confounds risk timing with risk level. Holding average exposure fixed separates the two, and shows that the regime layer earns its place through timing rather than through simply holding less.
- A reported null on regime-conditional signal fusion, together with the information coefficients that produce it, measured overall and within each state.
- A documented diagnosis of a failure mode in signal-to-allocation conversion: passing an unscaled indicator into a mean-variance objective allows the ratio of units between the linear and quadratic terms, rather than the information in the signal, to determine the solution, producing a degenerate single-asset portfolio.

1.5 Organisation

Section 2 reviews portfolio construction, regime-aware allocation and the treatment of textual sentiment, and positions this study against them. Section 3 specifies the framework. Section 4 reports the results, including robustness, ablation and the negative findings. Section 5 discusses implications and limitations, and Section 6 concludes.

2 Literature Review

2.1 *From Mean–Variance Optimisation to Risk-Based Allocation*

Modern portfolio construction begins with Markowitz (1952), who formalised allocation as a trade-off between expected return and variance. The framework is theoretically complete and practically fragile: its solutions are highly sensitive to the expected-return vector, and sample covariance matrices estimated from short windows on many assets are poorly conditioned in exactly the directions the optimiser exploits. DeMiguel et al. (2009) quantified the consequence, finding that a naive equal-weighted portfolio outperformed a range of optimised alternatives out of sample once estimation error was taken into account.

Two responses dominate. The first improves the inputs. Ledoit and Wolf (2004) introduced a shrinkage estimator that trades a small bias for a large variance reduction and yields a well-conditioned covariance matrix; the approach remains standard, and recent work extends shrinkage to higher moments (Lu and Tian, 2025). The second abandons expected returns altogether in favour of risk-based construction. Inverse-volatility weighting, risk parity and hierarchical risk parity (López de Prado, 2016) allocate on the basis of the risk model alone, and Asimit et al. (2025) develop the risk-budgeting formulation further by linking portfolio selection to risk sharing; Khodamoradi et al. (2023) pursue a related robustness objective through multi-interval mean-conditional-value-at-risk optimisation, and Kehinde et al. (2025) evaluate volatility targeting through an inverse data-envelopment formulation. Volatility management is a related idea applied at the portfolio level rather than the asset level: Moreira and Muir (2017) showed that scaling exposure inversely to recent volatility improves risk-adjusted returns across a range of factors, and Hood and Raughtigan (2025) examine how much of that improvement is attributable to trend following rather than to volatility management itself.

Table 1 positions the present study against these paradigms. This study sits between the two responses. It retains an expected-return term, but scales it explicitly so that its magnitude is commensurate with the risk term, and it separates the composition decision from the total-risk decision so that the two can be evaluated independently.

2.2 *Regime Dependence and Regime-Aware Allocation*

The premise that financial series alternate between statistically distinct states dates to Hamilton (1989), whose Markov-switching formulation remains the reference point. Ang and Bekaert (2002) applied regime switching to international allocation and found that correlations rise in bear states, weakening diversification precisely when it is most needed—an observation that motivates state-dependent covariance and state-dependent risk budgets alike. Kritzman et al. (2012) argued that regime awareness should change the strategy rather than only the commentary, and Barroso and Santa-Clara (2015) demonstrated a closely related mechanism at the factor level, showing that managing the volatility of a momentum strategy removes most of its crash risk.

Contemporary work continues along three lines. The first identifies regimes with increasingly flexible statistical machinery: Li et al. (2025) combine a jump model with allocation rules, Luo and Mulvey (2026) construct dual-regime signals with regime-dependent parameters, and Bukhari et al. (2026) make the state explicit inside a forecasting network by learning separate static and dynamic dependency graphs for each regime. The

Table 1 Portfolio construction paradigms and the position of this study

Paradigm	Representative work	Basis of allocation	Principal limitation
Mean–variance optimisation	Markowitz (1952)	Expected returns and covariance	Extreme sensitivity to estimated inputs
Naive diversification	DeMiguel et al. (2009)	None; equal weights	Ignores risk structure entirely
Shrinkage-based optimisation	Ledoit and Wolf (2004); Lu and Tian (2025)	Regularised covariance	Still requires expected returns
Risk-based construction	López de Prado (2016); Asimit et al. (2025)	Risk model only	Forgoes signal information by design
Volatility management	Moreira and Muir (2017); Hood and Raughtigan (2025)	Total risk scaled by recent volatility	Does not address composition
Regime-switching allocation	Ang and Bekaert (2002); Kritzman et al. (2012); Luo and Mulvey (2026)	State-dependent parameters	Component contributions rarely isolated
Learning-based allocation	Jiang et al. (2025); Ashrafzadeh et al. (2025); Agal et al. (2025)	Learned policy or predicted returns	Interpretability; frequently single-asset
This study	—	Regime-conditional signal, plus a relative regime risk budget	Signal layer tested and found not to contribute

second replaces discrete states with data-driven grouping or with directly modelled time-varying volatility; Kehinde et al. (2026) cluster assets dynamically and forecast within the resulting clusters, while Naidoo et al. (2025) estimate volatility as a mixed-frequency process driven by investor sentiment. The third embeds state-dependence inside a learning system, where the regime is implicit in the learned policy (Jiang et al., 2025). Across all three lines, the common gap is evidential rather than methodological. Regime-aware systems are usually evaluated against regime-unaware benchmarks in aggregate, so the reader learns that the system as a whole performs well but not which of its state-dependent components is responsible. The ablation design in Section 4 is intended to close that gap for the framework proposed here.

2.3 Machine Learning and Multimodal Signals

Machine learning entered this field early and has since become routine. Agent-based and reinforcement formulations were applied to daily equity trading by Lee et al. (2007), metaheuristic-optimised regression to price forecasting by Chou and Nguyen (2018), and pattern mining with fuzzy inference to trading-point detection by Huang et al. (2020); each improved predictive accuracy without addressing what the resulting forecasts imply for a portfolio. Gu et al. (2020) provide the benchmark comparative study in asset pricing, finding consistent gains from flexible models while noting that measured predictability is small. Applications to allocation are more recent: Ashrafzadeh et al. (2025) combine deep

learning with stock clustering for portfolio construction, Agal et al. (2025) apply learned risk estimates to asset allocation, Ha et al. (2026) compare machine-learning portfolios against a fixed all-weather allocation, and Singsiri and Yokrattanasak (2026) build portfolios directly on money-flow indicators of the kind used in this study. Reinforcement learning has been extended from single assets to multi-period, high-dimensional allocation (Jiang et al., 2025; Saijai and Pankwaen, 2026), and Wang et al. (2026) couple prediction with an explicit trading strategy.

Fusion of heterogeneous signals is a parallel development. Zhang et al. (2022) review decision fusion for equity prediction across a large body of primary studies and find that fused architectures usually outperform unimodal ones, while also documenting wide variation in fusion protocol and evaluation. Anbaee Farimani et al. (2024) propose an adaptive multimodal model that weights sources dynamically, and Zong et al. (2025) use gated cross-attention to stabilise multimodal fusion for movement prediction. What these studies establish is that fusion can help prediction. Whether a particular fusion rule helps a particular allocation decision is a separate question, and one that requires a component-level test rather than an aggregate comparison.

2.4 *Textual Sentiment and Market-Derived Proxies*

Textual sentiment entered finance through dictionary methods, and Loughran and McDonald (2011) demonstrated that general-purpose sentiment lexicons misclassify financial language badly enough to require domain-specific alternatives. Transformer models and large language models have since displaced lexicons: Officioso et al. (2026) combine language-model sentiment with market data, Muhammad et al. (2026) benchmark large language models for target-based financial sentiment, Mun and Kim (2025) derive investment signals from language-model sentiment, and Chuang et al. (2025) compare language models on financial news for sentiment and price prediction. Collectively these studies indicate that genuine textual sentiment carries information beyond price history. George and Murugesan (2025), in this journal, illustrate what extracting it actually involves: a corpus of more than twenty thousand stock-related messages, five feature-extraction schemes and fourteen classifiers, evaluated as a classification problem in its own right before any allocation question is reached.

They also indicate what such a study requires: a time-stamped corpus with coverage of the traded instruments. That requirement is binding. For broad exchange-traded funds there is no instrument-level news corpus of comparable coverage, which is why studies of this kind are almost always conducted on individual equities. The present study therefore uses market-derived proxies for attention and stress rather than textual sentiment, states this explicitly in Section 3.3, and—rather than assuming the proxies are adequate—measures their predictive content directly and reports that they are not.

2.5 *Evaluation Practice*

A final strand concerns whether reported results should be believed. Harvey and Liu (2015) show that conventional significance thresholds are inadequate once the number of strategies examined is taken into account, and Valls and Steurer (2026) document the same problem specifically for volatility-based strategies, while Sheppert (2026) proposes an objective function designed to resist it. Zhang et al. (2025) make the related point that the model selected as optimal in sample is frequently not the one that performs best out of it. Ledoit

and Wolf (2008) provide robust inference for Sharpe ratio comparisons under non-normal, autocorrelated returns, and Politis and Romano (1994) supply the stationary block bootstrap used here to construct interval estimates that respect serial dependence. Newey and West (1987) remains the standard correction for autocorrelated and heteroskedastic test statistics. Raimundo and Bravo (2026) survey the broader intersection of forecasting and portfolio theory and reach a similar conclusion about the weight placed on point estimates.

This study adopts these practices as design constraints rather than as afterthoughts: every specification examined is reported, inference is corrected for serial dependence, interval estimates accompany point estimates, and the comparison against a costless daily-rebalanced equal-weighted benchmark is deliberately unfavourable to the proposed framework.

3 Proposed Methodology

This section specifies the framework in full: the data, the exact form of every input feature, the regime model, the fusion rule, the optimisation problem, and the walk-forward protocol under which it is evaluated. The level of detail is deliberate, so that the study can be reproduced independently from this description alone.

3.1 Data and Study Universe

The primary study universe consists of ten exchange-traded funds selected to span the major liquid asset classes available to a diversified allocator: US large capitalisation equity (SPY), US large capitalisation growth equity (QQQ), US small capitalisation equity (IWM), developed international equity (EFA), emerging market equity (EEM), long-maturity US Treasuries (TLT), intermediate-maturity US Treasuries (IEF), investment-grade corporate credit (LQD), gold (GLD) and US real estate (VNQ). Breadth matters for the research question. A framework whose stated purpose is to redirect risk across market states can only be tested where there is somewhere else for risk to go; a single-sector universe offers no such destination, a point returned to in Section 4.12.

Daily open, high, low, close and volume series are obtained from Yahoo Finance through the `yfinance` interface, adjusted for splits and dividends. The download window is 1 January 2007 to 27 August 2026. The end date is not a modelling choice: it is every session available at the date the analysis was run, so the sample boundary carries no information about the results. Indicator warm-up and the 252-session covariance estimation window consume the first year, so the evaluated out-of-sample period runs from January 2008 to August 2026 and covers the global financial crisis, the 2011 sovereign debt episode, the 2020 pandemic crash, the 2022 inflation-driven bear market and the subsequent recovery.

Sessions from 1 January 2024 onward are treated as a holdout. That block postdates the design of the framework and every specification decision reported here; none of it was examined while the model was being built. It enters the full-sample results for completeness and is reported separately in Section 4.2, where it serves as an out-of-sample check rather than as evidence used in fitting. A secondary universe of ten mega-capitalisation US equities over 2018–2026 is carried through the same pipeline and reported separately in Section 4.12.

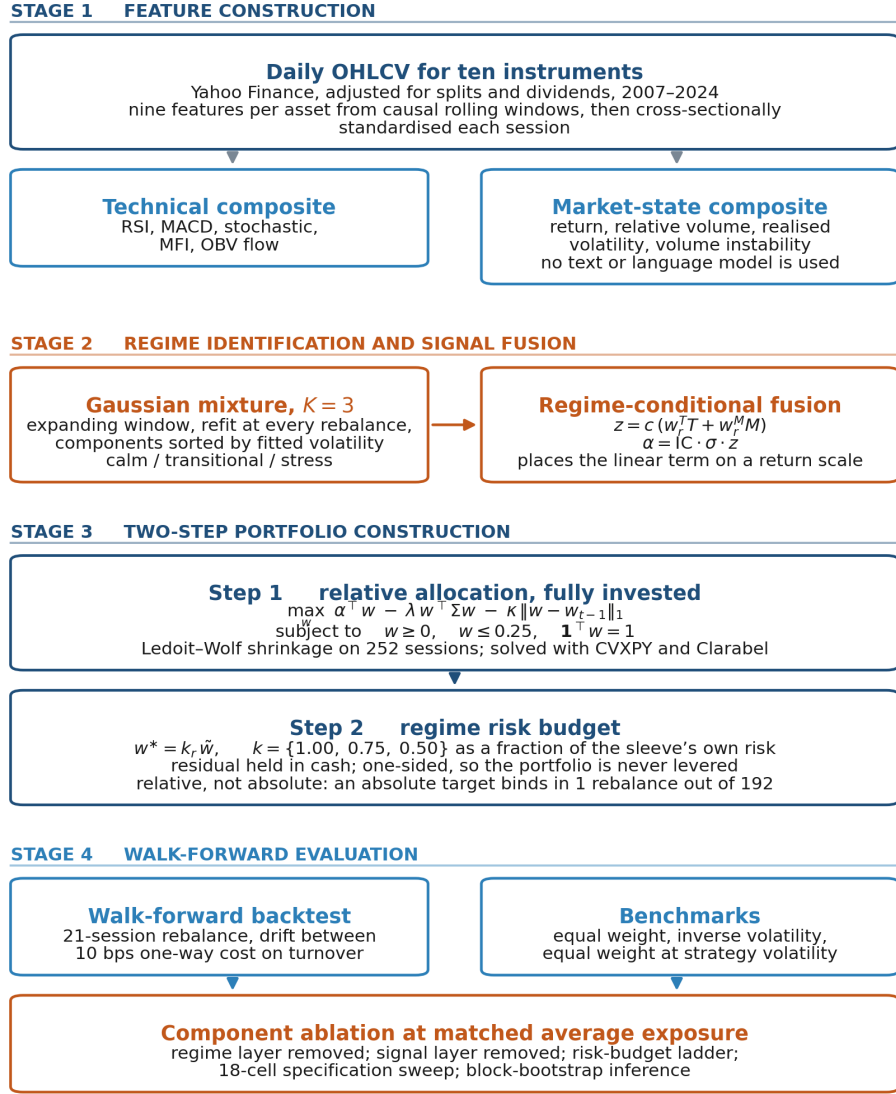


Figure 1 The proposed framework. Stage 1 constructs causal, cross-sectionally standardised features; stage 2 identifies the market state and fuses the two composites; stage 3 separates relative composition from total risk; stage 4 evaluates the result and ablates its components.

3.2 Feature Construction

Nine features are computed for each asset on each session. All are causal: every value at session t uses only information available at the close of session t . Table 2 gives the exact definition of each, and no other input enters the framework.

Table 2 Input features. C_t , H_t , L_t and V_t denote close, high, low and volume. All features are computed per asset, then cross-sectionally standardised.

Feature	Composite	Definition	Window
RSI	Technical	Relative strength index, centred: $RSI_t - 50$	14
MACD	Technical	Moving average convergence divergence line, scaled by price: $MACD_t / C_t$	12, 26, 9
Stochastic	Technical	Stochastic oscillator %K, centred: $\%K_t - 50$	14, smooth 3
MFI	Technical	Money flow index, centred: $MFI_t - 50$	14
OBV flow	Technical	Twenty-session change in on-balance volume, normalised by average volume: $(OBV_t - OBV_{t-20}) / \bar{V}_{t,20}$	20
Return	Market state	Five-session simple return: $C_t / C_{t-5} - 1$	5
Relative volume	Market state	$V_t / \bar{V}_{t,20}$	20
Realised volatility	Market state	Annualised standard deviation of daily returns, entered with a negative sign	20
Volume instability	Market state	Standard deviation of daily volume changes, entered with a negative sign	20

Each feature is standardised cross-sectionally on every session, so that a value expresses how an asset ranks against its peers that day rather than against its own history. Writing $x_{i,t}$ for the raw feature of asset i and N for the number of assets,

$$z_{i,t} = \frac{x_{i,t} - \bar{x}_t}{s_t}, \quad \bar{x}_t = \frac{1}{N} \sum_j x_{j,t}, \quad s_t = \text{sd}_j(x_{j,t}). \quad (1)$$

Cross-sectional standardisation is used rather than time-series standardisation because the allocation problem is a relative one: the optimiser must decide which assets to prefer today, not whether today is unusual by historical standards. It also removes the common market factor from the scores, so that a market-wide move does not appear as a signal on every asset simultaneously.

The standardised features are then averaged into two composites. The technical composite is the equally weighted mean of the five technical features. The market-state composite is the equally weighted mean of the four market-state features, taken with the signs given in Table 2, so that recent strength and active participation raise the score while realised volatility and unstable volume lower it.

3.3 What the Second Composite Is, and What It Is Not

The second composite is constructed entirely from price and volume. No news article, social media post, corporate announcement or any other text is read at any point in this study, and no lexicon, classifier or language model is applied. The composite is a market-derived proxy for the same latent quantities that textual sentiment measures aim to capture—attention, participation and stress—and it is reported here as exactly that.

This is stated plainly because the distinction is material to how the results should be read. Proxy construction of this kind is common in the literature and inexpensive to compute, but it cannot be credited with the contextual information a transformer model extracts from

financial text, and it should not be described as though it could. Section 4.8 measures the predictive content of both composites directly and reports the outcome, including where that outcome is unfavourable. Extending the framework to genuine textual sentiment requires a time-stamped news corpus with coverage of the traded universe; for broad exchange-traded funds no such corpus exists at instrument level, which is a limitation of the design rather than an oversight, and Section 5 sets out what would be needed to remove it.

3.4 *Regime Identification*

Market states are identified with a Gaussian mixture model over two features, each computed as a cross-sectional average across the universe: the twenty-session rolling standard deviation of daily returns, and the ten-session rolling mean of daily returns. The first captures the volatility of the environment, the second its direction. Both are standardised before fitting.

The mixture uses $K = 3$ components with full covariance matrices, three random restarts and a fixed seed. It is refitted at every rebalance date on an expanding window containing only sessions strictly earlier than that date, with a minimum of 252 observations; the state of the current session is then obtained by prediction from that fit. No observation ever contributes to the model that labels it.

One implementation detail is essential and is easy to overlook. The component indices returned by a mixture fit are arbitrary and permute freely between refits, so labels from consecutive fits are not comparable unless they are anchored. Components are therefore sorted in ascending order of fitted mean volatility before use, which makes regime 0 the calm state, regime 1 the transitional state and regime 2 the stress state consistently across every refit. Omitting this step does not produce an error; it produces a silently scrambled regime series and, if the resulting labels are compared across fits, a spurious appearance of instability.

The choice $K = 3$ is not selected by information criterion. As Section 4.10 reports, both the Akaike information criterion (AIC) and the Bayesian information criterion (BIC) continue to fall as components are added over the range tested, which is the usual behaviour of mixture models on persistent financial series and reflects their tendency to absorb heavy tails with additional components. The choice is instead made on persistence and interpretability: three components yield states that last long enough to act on and that admit a clear economic reading, whereas finer partitions fragment into short-lived states that a monthly rebalancing schedule cannot exploit. This is a modelling judgement and is reported as such.

3.5 *Regime-Conditional Signal Fusion*

The two composites are combined into a single score using weights that depend on the prevailing regime, together with a per-asset confidence term. Let $T_{i,t}$ and $M_{i,t}$ denote the standardised technical and market-state composites for asset i , and let $r_t \in \{0, 1, 2\}$ be the regime. The fused score is

$$z_{i,t} = c_{i,t} [w_{r_t}^T T_{i,t} + w_{r_t}^M M_{i,t}], \quad (2)$$

with regime weights (w^T, w^M) of $(0.7, 0.3)$ in the calm state, $(0.5, 0.5)$ in the transitional state and $(0.3, 0.7)$ in the stress state. The direction of this schedule follows the hypothesis stated in Section 1, that trend-following information is more reliable in quiet trending

markets while attention and stress information becomes relatively more informative when markets are disturbed. Section 4.8 tests that hypothesis rather than assuming it.

The confidence term $c_{i,t}$ down-weights assets whose two composites disagree:

$$c_{i,t} = \text{clip}\left(0.5 + 0.5 \rho_{i,t}^{(63)}, 0.1, 1.0\right), \quad (3)$$

where $\rho_{i,t}^{(63)}$ is the trailing 63-session correlation between the two composites for that asset. Agreement between independent views is treated as evidence; disagreement is treated as noise and shrinks the score toward zero.

The fused score is converted into an expected excess return using the standard information-ratio scaling of Grinold and Kahn (2000), which places the score on a return scale rather than leaving it as a dimensionless quantity:

$$\alpha_{i,t} = \text{IC} \cdot \sigma_{i,t} \cdot z_{i,t}, \quad (4)$$

where $\sigma_{i,t}$ is the annualised volatility of asset i from the current covariance estimate and $\text{IC} = 0.05$ is an assumed information coefficient (IC).

Equation (4) is not a cosmetic step. If a raw indicator is passed to a mean-variance objective directly, the units of the linear term and of the quadratic risk term are unrelated, and their ratio—not the information in the signal—decides the solution. An indicator such as on-balance volume takes values of order 10^8 to 10^9 , whereas annualised variances are of order 10^{-2} ; the linear term then dominates by ten orders of magnitude and the optimiser degenerates to a corner solution, holding a single asset regardless of the risk model. Section 4.9 documents this failure mode explicitly, because it is the source of a behaviour visible in the originally submitted results.

3.6 Portfolio Construction

Allocation proceeds in two steps, which separate two decisions that are often conflated: which assets to hold relative to one another, and how much total risk to carry.

Step 1: relative allocation.

The risky sleeve solves a quadratic program with a linear transaction-cost penalty,

$$\max_w \alpha_t^\top w - \lambda w^\top \Sigma_t w - \kappa \|w - w_{t-1}\|_1 \quad (5)$$

$$\text{s.t. } w \geq 0, \quad w \leq w_{\max}, \quad \mathbf{1}^\top w = 1, \quad (6)$$

where Σ_t is the annualised covariance matrix, $\lambda = 2$ is the risk aversion coefficient, $w_{\max} = 0.25$ caps any single position at one quarter of the sleeve, and w_{t-1} is the weight vector carried in from the previous rebalance. The turnover penalty $\kappa = c \cdot 252/h$ annualises a one-way cost of $c = 10$ basis points over a rebalancing interval of $h = 21$ sessions, so that the cost of trading is expressed on the same annual basis as the return and risk terms.

The equality constraint $\mathbf{1}^\top w = 1$ is what makes this step purely relative. It fixes the sleeve to be fully invested, so the optimiser can only express a view about composition, never about how much risk to take. That second decision is made explicitly in step 2 instead of emerging as a by-product of the constraint set.

Σ_t is estimated by Ledoit–Wolf shrinkage (Ledoit and Wolf, 2004) on the trailing 252 daily returns and annualised by a factor of 252. Shrinkage is used because a sample covariance matrix estimated from 252 observations on ten assets is poorly conditioned, and a mean-variance optimiser will concentrate weight precisely on the directions in which the estimate is least reliable. The problem is convex and is solved with CVXPY using the Clarabel interior-point solver.

Step 2: regime risk budget.

The sleeve is then scaled by a factor that depends only on the regime:

$$w_t^* = k_{r_t} \tilde{w}_t, \quad k = \{ \text{calm} : 1.00, \text{transitional} : 0.75, \text{stress} : 0.50 \}, \quad (7)$$

with the residual $1 - k_{r_t}$ held in cash, which is assumed to earn zero. Scaling is one-sided: the framework de-risks in disturbed states but never levers up in calm ones, so any performance difference is attributable to risk reduction rather than to leverage.

The budget in Equation (7) is expressed as a fraction of the sleeve’s own risk rather than as an absolute volatility target, and this choice is consequential. An absolute target de-risks only when the sleeve’s estimated volatility exceeds the target level, so its effect depends on the volatility of the universe it is applied to. Targets calibrated for a concentrated equity portfolio—of the order of 10 to 20 per cent annualised—are simply never reached by a diversified ten-asset sleeve whose own realised volatility is close to 7 per cent. Under those targets the constraint binds in one rebalance out of 224 in this study, and the regime layer is inert: it is present in the specification, consumes computation, and does nothing. A relative budget is scale-free by construction and remains active in any universe. Section 4.6 reports both variants and the difference between them.

3.7 Walk-Forward Protocol

The framework is rebalanced every 21 sessions. Between rebalances positions are held and allowed to drift with realised returns, so that reported turnover reflects trading actually undertaken rather than a notional daily reweighting. One-way transaction costs of 10 basis points are charged on the traded fraction $\|w_t^* - w_{t-1}\|_1$ on the rebalance session; Section 4.9 varies this assumption from 0 to 25 basis points.

Every estimated quantity uses only data preceding the decision it informs: the covariance matrix from the trailing 252 sessions before the rebalance date, the mixture model from an expanding window strictly earlier than that date, and all features from causal rolling windows. No parameter is tuned on the evaluation period; the configuration reported as primary is the one specified above, and Section 4.3 reports the full grid of alternatives rather than the best cell of it.

3.8 Evaluation Design

Performance is summarised by compound annual growth rate (CAGR), annualised volatility, Sharpe ratio computed at a zero risk-free rate, maximum drawdown (Max DD) and Calmar ratio. Three benchmarks are used. The first is an equal-weighted portfolio of the same ten assets, rebalanced daily and charged no transaction costs, which is deliberately a generous comparator. The second is an inverse-volatility portfolio using trailing 60-session standard deviations, also costless, which isolates whether the framework adds anything beyond

elementary risk weighting. The third rescales the equal-weighted portfolio to the strategy's realised volatility, which removes the mechanical advantage a lower-risk portfolio enjoys in ratio comparisons.

Statistical inference on daily excess returns uses a paired t -test, a Newey–West t -statistic with ten lags to accommodate autocorrelation and heteroskedasticity, and a stationary block bootstrap with 21-session blocks and 2,000 replications for interval estimates of the Sharpe ratio and of the Sharpe difference. Component contributions are isolated by ablation: removing the regime layer while holding average exposure fixed, removing it while holding full exposure, and removing the signal layer entirely. Robustness is assessed over a grid of random seeds, covariance estimation windows and rebalancing frequencies, with every cell reported.

4 Results

All results in this section come from the single specification set out in Section 3. No parameter was selected by reference to the evaluation period, and every specification examined is reported, including those that are unfavourable to the framework.

4.1 Headline Performance

Table 3 reports the primary comparison over the 4,689 trading sessions from 7 January 2008 to 26 August 2026.

Table 3 Primary results, ten-asset universe, January 2008 to August 2026. Sharpe ratios use a zero risk-free rate and are computed as compound annual growth divided by annualised volatility. Benchmarks are rebalanced daily and charged no transaction costs.

	CAGR	Volatility	Sharpe	Max DD	Calmar
Regime-conditional strategy	5.65%	6.44%	0.878	−18.98%	0.298
Equal-weight benchmark	8.45%	13.45%	0.628	−36.08%	0.234
Inverse-volatility benchmark	7.46%	8.78%	0.849	−24.09%	0.309
Equal-weight at strategy volatility	4.19%	6.44%	0.651	−18.69%	0.224

Three readings of Table 3 are warranted, and they do not all favour the framework.

Against the equal-weighted benchmark the framework improves the Sharpe ratio from 0.628 to 0.878, a gain of 40 per cent, and reduces maximum drawdown from −36.08 per cent to −18.98 per cent, a reduction of 47 per cent. It does so while earning a lower compound return, 5.65 per cent against 8.45 per cent per annum. The framework is not a return-enhancing strategy and is not presented as one; it is a risk-reduction strategy whose return sacrifice is smaller than its risk reduction.

Against the volatility-matched equal-weighted benchmark, which removes the mechanical advantage a lower-risk portfolio enjoys in any ratio comparison, the framework retains a Sharpe advantage of 0.878 against 0.651, but loses on maximum drawdown, −18.98 per cent against −18.69 per cent. Deleveraging a diversified benchmark is therefore a competitive way to reduce worst-case loss, and the framework's drawdown advantage over the unlevered benchmark should not be read as evidence that it dominates every risk-reduced alternative.

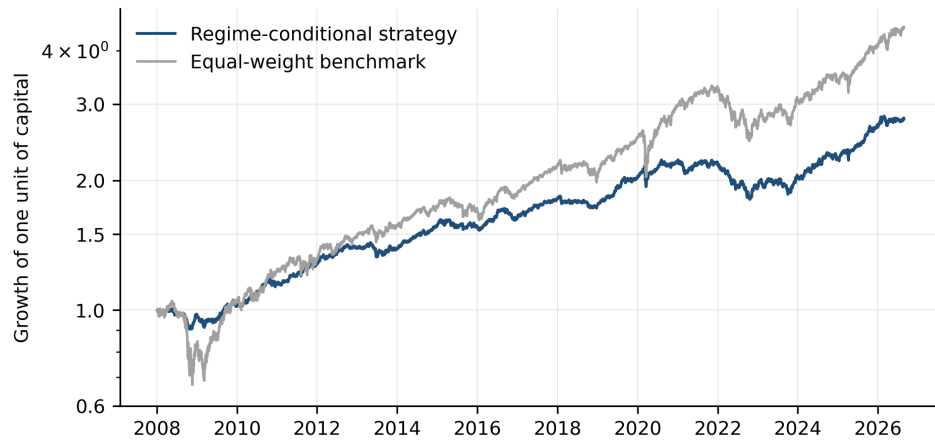


Figure 2 Cumulative growth of one unit of capital, logarithmic scale

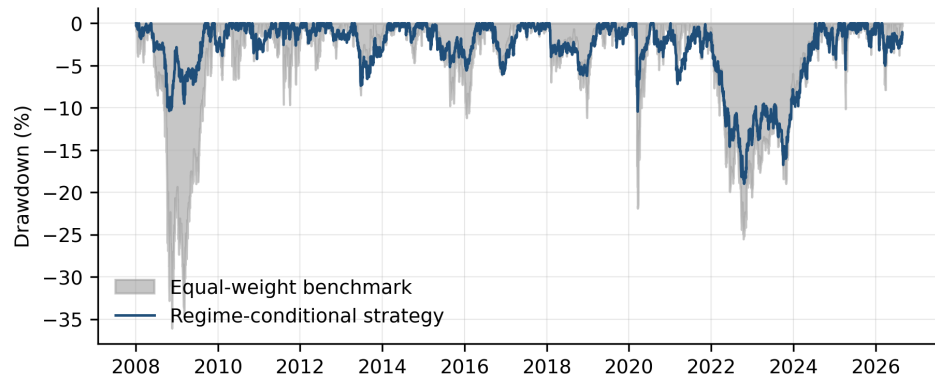


Figure 3 Drawdown from running peak

Against the inverse-volatility benchmark the comparison is mixed. The framework holds a higher Sharpe ratio, 0.878 against 0.849, and a materially smaller maximum drawdown, -18.98 per cent against -24.09 per cent, but a lower Calmar ratio, 0.298 against 0.309, because inverse-volatility weighting compounds faster. Elementary risk weighting remains a strong comparator on a diversified multi-asset universe, and the margin over it is narrow enough that it should be read as a tie on risk-adjusted return rather than as a win.

Figure 2 shows the cumulative growth of one unit of capital and Figure 3 the drawdown profile of each series.

The two figures make the shape of the result visible. The framework's equity curve tracks below the benchmark through extended advances—most conspicuously 2013, 2017, 2021 and 2023—and separates upward during 2008, 2011, 2020 and 2022. The drawdown plot shows the same asymmetry more directly: the benchmark reaches -36 per cent in 2008 and -25 per cent in 2022, while the framework's deepest excursions are approximately half as large. Recovery is correspondingly faster, since a shallower loss requires a smaller subsequent gain to restore the previous peak.

4.2 Out-of-Sample Holdout, 2024–2026

Every specification decision in this study—the universe, the feature set, the number of regimes, the risk budget, the rebalancing interval and the cost assumption—was fixed before any session after December 2023 was examined. The 665 sessions from January 2024 to August 2026 therefore function as a holdout rather than as a further robustness check on the same data. Table 4 reports both blocks side by side.

Table 4 Performance before and after the holdout boundary. The two blocks are disjoint; the full-sample results of Table 3 pool them.

	2008–2023 ($n = 4,024$)		2024–2026 ($n = 665$)	
	Sharpe	Max DD	Sharpe	Max DD
Regime-conditional strategy	0.738	−18.98%	1.614	−5.48%
Equal-weight benchmark	0.532	−36.08%	1.391	−10.13%
Inverse-volatility benchmark	0.764	−24.09%	1.397	−6.70%
Equal-weight at strategy volatility	0.560	−17.66%	1.378	−7.04%
No regime timing, full exposure	0.785	−22.53%	1.710	−6.47%

In the holdout the framework records a higher Sharpe ratio than all three benchmarks, 1.614 against 1.391, 1.397 and 1.378, and a smaller maximum drawdown than all three, −5.48 per cent against −10.13, −6.70 and −7.04 per cent. Two of those comparisons reverse the in-sample ordering: the inverse-volatility benchmark and the volatility-matched equal-weighted benchmark both beat the framework on Sharpe ratio before 2024 and both trail it afterwards. The deepest equal-weighted drawdown in the holdout, the 34-session decline from 20 February to 8 April 2025, returned −10.05% for the benchmark against −4.63% for the framework, an advantage of 5.42 percentage points that is consistent with the four earlier episodes in Table 6.

Three qualifications limit what this establishes. First, 665 sessions are too few to settle anything on their own: the bootstrap interval for the Sharpe difference in the holdout is $[-0.352, 0.742]$, wider than the full-sample interval and comfortably containing zero, and the framework again earns less in absolute terms, giving up 3.07 percentage points of annual return ($p = 0.348$). Second, the holdout contains no episode comparable to 2008 or 2020; every series in it posts a Sharpe ratio above 1.3, so the period is benign and the relevant information is the ordering rather than the level. Third, the ablation ordering is unchanged—the variant with no regime timing at full exposure again records the highest Sharpe ratio, 1.710—so the holdout does not rescue any claim that Section 4.6 rejects.

What the holdout does establish is narrower and still worth stating. The framework’s defensive behaviour is not an artefact of the sample on which it was specified. It was built without reference to these sessions, and on them it continues to reduce drawdown against every benchmark considered while remaining competitive on risk-adjusted return.

4.3 Robustness Across Specifications

The primary configuration fixes a random seed, a covariance estimation window and a rebalancing interval. Table 5 reports every combination of three seeds, two estimation windows and three rebalancing intervals, so that the reported result can be located within the distribution of specifications rather than presented in isolation.

Table 5 Specification sweep. All eighteen cells reported; the last two columns record whether the cell beats the equal-weight benchmark on each measure. Sharpe ratios here are the annualised arithmetic mean over volatility, so that all eighteen cells are computed identically; on the same basis the equal-weight benchmark attains 0.670 with a maximum drawdown of -36.1% .

Seed	Cov. window	Rebalance	Sharpe	Max DD	Sharpe win	DD win
0	126	10	0.868	-20.7%	yes	yes
0	126	21	0.954	-19.6%	yes	yes
0	126	42	0.924	-19.7%	yes	yes
0	252	10	0.823	-20.8%	yes	yes
0	252	21	0.905	-19.0%	yes	yes
0	252	42	0.899	-20.4%	yes	yes
7	126	10	0.881	-20.6%	yes	yes
7	126	21	0.956	-19.6%	yes	yes
7	126	42	0.912	-19.6%	yes	yes
7	252	10	0.818	-21.1%	yes	yes
7	252	21	0.891	-19.0%	yes	yes
7	252	42	0.888	-20.4%	yes	yes
42	126	10	0.853	-20.6%	yes	yes
42	126	21	0.928	-19.6%	yes	yes
42	126	42	0.893	-19.7%	yes	yes
42	252	10	0.813	-21.1%	yes	yes
42	252	21	0.887	-19.0%	yes	yes
42	252	42	0.853	-20.4%	yes	yes

Every one of the eighteen specifications beats the equal-weighted benchmark on both the Sharpe ratio and maximum drawdown. The Sharpe ratio ranges from 0.813 to 0.956 and the maximum drawdown from -21.1 to -19.0 per cent. The primary configuration appears in this table at 0.887, which is the median of the eighteen values: it is not the best cell, and the best cell exceeds it by 0.069. Sensitivity to the random seed is small, which indicates that the mixture fit is stable; sensitivity to the rebalancing interval is larger, with ten-session rebalancing consistently weakest because it incurs more turnover for no additional information.

The same sweep is unfavourable on one comparison. In none of the eighteen cells does the framework beat the volatility-matched equal-weighted benchmark on maximum drawdown. This is a consistent finding rather than an artefact of the primary configuration, and it bounds the drawdown claim precisely: the framework reduces drawdown relative to an unlevered diversified benchmark, not relative to every portfolio held at the same volatility.

4.4 Behaviour in Stress Episodes

Table 6 reports cumulative returns over four episodes chosen before the analysis on the basis of documented market history rather than by inspection of the results.

The framework outperforms in all four episodes, by between 7.7 and 22.7 percentage points. All four predate the holdout boundary; the deepest benchmark decline after it, analysed in Section 4.2, produced an advantage of 5.4 percentage points. The mechanism is visible in the exposure series: the mixture model classifies these periods into the transitional or stress state, and the risk budget reduces invested exposure accordingly. The margin is widest in 2008, the longest and deepest of the four, and narrowest in 2022, which is consistent

Table 6 Cumulative return over stress episodes. Differences are stated in percentage points (pp).

Episode	Sessions	Strategy	Equal-weight	Difference
Global financial crisis, Sep 2008–Mar 2009	130	−7.05%	−29.78%	+22.74 pp
Euro area crisis, Jul–Oct 2011	65	+3.41%	−7.24%	+10.65 pp
COVID-19 crash, Feb–Mar 2020	24	−6.29%	−21.28%	+14.99 pp
Inflation bear market, Jan–Oct 2022	196	−16.76%	−24.49%	+7.73 pp

with the design—a gradual, correlated decline across both equities and bonds offers less scope for a de-risking rule than an abrupt volatility shock does.

Complementing the episode analysis, the framework’s average return on the worst 5 per cent of benchmark sessions ($n = 234$) is −0.59 per cent against −2.01 per cent for the benchmark, a cushion of 1.42 percentage points per session. Downside capture is 0.38 and upside capture 0.41, giving a capture ratio of 0.93. The ratio being close to unity is informative: the framework participates less in both directions in roughly equal proportion, so its advantage comes predominantly from carrying less risk in adverse states rather than from any directional skill.

4.5 Year-by-Year Performance

Table 7 reports every calendar year in the sample, with both wins and losses shown.

Table 7 Calendar year performance. Sharpe and maximum drawdown computed within each year. 2026 covers January to August only; the sample ends on 26 August 2026.

Year	Strategy	Equal-weight	Strat. Sharpe	EW Sharpe	Sharpe win	DD win
2008	−2.28%	−16.32%	−0.33	−0.47	yes	yes
2009	+4.36%	+24.91%	0.60	1.18	no	yes
2010	+13.75%	+18.31%	2.05	1.40	yes	yes
2011	+10.36%	+5.75%	1.73	0.45	yes	yes
2012	+8.91%	+13.49%	1.96	1.58	yes	yes
2013	−0.58%	+6.17%	−0.06	0.72	no	yes
2014	+13.03%	+9.97%	2.94	1.49	yes	yes
2015	−0.48%	−1.56%	−0.05	−0.13	yes	yes
2016	+4.85%	+8.49%	0.94	0.97	no	yes
2017	+10.92%	+16.59%	2.65	3.15	no	yes
2018	−3.51%	−5.26%	−0.78	−0.52	no	yes
2019	+15.72%	+22.55%	2.84	3.02	no	yes
2020	+9.63%	+19.32%	1.23	0.96	yes	yes
2021	−0.78%	+10.15%	−0.08	1.12	no	no
2022	−12.99%	−19.30%	−1.55	−1.23	no	yes
2023	+6.83%	+16.53%	0.90	1.52	no	yes
2024	+9.38%	+9.58%	1.25	1.00	yes	yes
2025	+18.53%	+20.17%	2.24	1.67	yes	yes
2026	+4.80%	+10.93%	0.96	1.34	no	yes

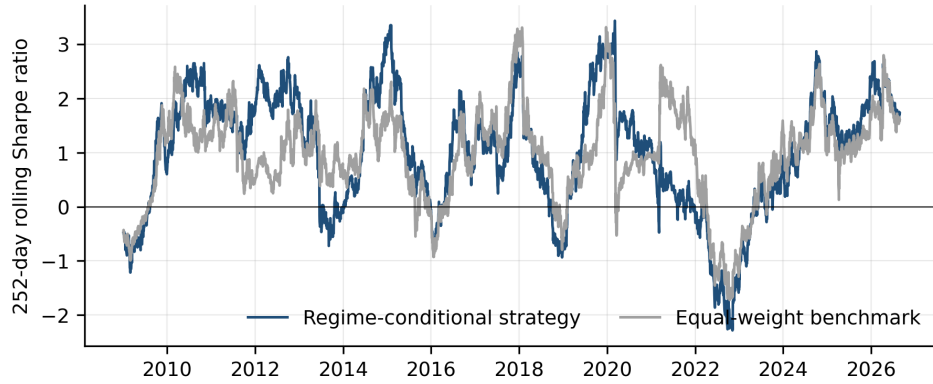


Figure 4 Rolling 252-session Sharpe ratio

The framework wins on Sharpe ratio in 9 of 19 years and on drawdown in 18 of 19. This distribution is the honest summary of the strategy: it loses on risk-adjusted return in most individual years, particularly strong advancing years such as 2009, 2017, 2021 and 2023, and recovers the deficit in the minority of years when the benchmark suffers. The three years after the holdout boundary follow the same pattern: drawdown wins in all three, Sharpe wins in two. Two years merit comment. In 2021 the framework loses on both measures, the only such year, because the mixture model spent much of that year in the transitional state during what turned out to be a sustained advance. In 2018 the framework loses on Sharpe despite a smaller loss in absolute terms, an arithmetic consequence of a smaller negative return divided by a much smaller volatility.

Figure 4 shows the rolling one-year Sharpe ratio of both series. The framework's rolling Sharpe is positive in 81.9 per cent of windows against 85.8 per cent for the benchmark, and exceeds the benchmark in 53.4 per cent of windows. Its advantage is therefore episodic rather than persistent, which is what Table 7 leads one to expect.

4.6 Component Ablation

The framework has two active layers, signal fusion and the regime risk budget. Table 8 removes each in turn. The comparison requires care, because scaling a portfolio toward cash changes its return and its volatility together; comparing a de-risked strategy against a fully invested variant therefore confounds the timing of risk with its level. The first ablation holds average invested exposure fixed at the framework's own 92.1 per cent and applies it constantly, so that only the timing differs.

The first comparison is the one the framework is built on, and it succeeds. Holding average exposure fixed, allocating that exposure according to the regime rather than uniformly raises the Sharpe ratio from 0.795 to 0.878 and improves maximum drawdown from -22.26 to -18.98 per cent. The difference in mean return is not significant ($p = 0.695$), which is the expected signature of a pure risk-timing effect: the same average risk is carried, but it is carried at better moments. This is the only comparison in the study in which the framework improves both measures at once, and it is the claim the paper rests on.

The second comparison bounds the cost of that timing. Against a fully invested variant, the framework gives up approximately 1.3 percentage points of annual return, and the difference is statistically significant ($p = 0.016$). De-risking is not free, and its Sharpe

Table 8 Component ablation. The matched-exposure control holds the same average invested exposure as the full framework but applies it without regard to the regime.

Configuration	CAGR	Sharpe	Max DD	Exposure	<i>p</i>
Full framework	5.65%	0.878	−18.98%	92.1%	—
No regime timing, same average exposure	5.43%	0.795	−22.26%	92.1%	0.695
No regime timing, full exposure	6.91%	0.923	−22.53%	100.0%	0.016
No signal layer	5.75%	0.908	−18.87%	92.1%	0.650
Equal-weight benchmark	8.45%	0.628	−36.08%	100.0%	—

ratio under this comparison is lower, 0.878 against 0.923. What the framework buys with that return is 3.55 percentage points of maximum drawdown. Whether that exchange is worthwhile is a preference, not a fact, and the paper reports both sides of it.

The third comparison is a null, and it concerns the layer the original framework was named for. Removing the signal fusion layer entirely—setting expected returns to zero so that the optimiser becomes a constrained minimum-variance problem—changes the Sharpe ratio from 0.878 to 0.908 and the maximum drawdown from −18.98 to −18.87 per cent. Both movements favour removal, and neither is statistically significant ($p = 0.650$). The signal fusion layer contributes nothing detectable to portfolio outcomes in this universe, and its point estimate is mildly adverse. Section 4.8 establishes why.

4.7 Form of the Regime Risk Budget

Table 9 varies the risk budget itself, including the absolute-target formulation used in the framework as originally specified.

Table 9 Risk budget specifications. Relative budgets are stated as the fraction of the sleeve's own volatility retained in the calm, transitional and stress states.

Budget	CAGR	Sharpe	Max DD	Exposure
Relative 1.00 / 0.75 / 0.50 (primary)	5.65%	0.878	−18.98%	92.1%
Relative 1.00 / 0.80 / 0.60	5.77%	0.871	−19.95%	93.7%
Relative 1.00 / 0.60 / 0.30	5.31%	0.881	−16.62%	87.6%
Relative 1.00 / 0.50 / 0.25	5.21%	0.891	−15.71%	84.9%
Absolute targets, 20 / 15 / 10% annualised	6.93%	0.923	−22.86%	100.0%

Two findings follow. First, the Sharpe ratio varies little across the four relative budgets, from 0.871 to 0.891, while maximum drawdown improves monotonically as the budget tightens, from −19.95 to −15.71 per cent. The result is therefore not an artefact of the particular multipliers chosen; the mildest of the four is reported as primary precisely because it is the least aggressive, and the tighter budgets would have produced a marginally better headline on both measures.

Second, the absolute-target formulation is inert. Its mean invested exposure is 99.99 per cent, and the constraint binds in one rebalance out of 224. Targets of 10 to 20 per cent annualised volatility are never reached by a sleeve whose own realised volatility is 6.4 per cent, so under that formulation the regime layer is present in the specification but has no effect on the portfolio. Its apparently superior Sharpe ratio of 0.923 is simply the fully invested portfolio of Table 8, reached by a different route. This is the practical reason the

relative formulation is adopted: an absolute volatility target is not scale-free, and a target calibrated for one universe silently disables the mechanism in another.

4.8 Predictive Content of the Signals

Table 10 measures the predictive content of the two composites directly, as the cross-sectional Spearman information coefficient between the score on session t and the return on session $t + 1$, with Newey–West t -statistics.

Table 10 Information coefficients against next-session cross-sectional returns. Newey–West t -statistics with ten lags in parentheses.

	Technical	Market state	Fused
All sessions ($n = 4,669$)	−0.030 (−2.14)	−0.026 (−2.26)	−0.033 (−2.42)
Calm regime ($n = 3,339$)	−0.042 (−2.48)	−0.040 (−3.02)	—
Transitional regime ($n = 1,183$)	−0.012 (−0.48)	+0.007 (+0.31)	—
Stress regime ($n = 147$)	+0.087 (+1.30)	+0.040 (+0.66)	—
2008–2023 ($n = 4,025$)	−0.035 (−2.31)	−0.032 (−2.67)	−0.039 (−2.73)
2024–2026 holdout ($n = 644$)	+0.001 (+0.03)	+0.013 (+0.37)	+0.009 (+0.23)

Over the full sample the information coefficients are negative and statistically significant for both composites and for their fusion. A composite whose cross-sectional score is negatively related to next-session returns cannot improve an allocation that uses it in the stated direction, and the ablation in Table 8 is the portfolio-level consequence of the numbers in Table 10.

The regime breakdown does not support the fusion hypothesis. Section 3.5 assigns greater weight to the market-state composite in the stress regime on the hypothesis that it becomes relatively more informative there. The point estimates run the other way: in the stress regime the technical composite reads higher than the market-state composite, and in the calm regime the two are indistinguishable. The contrast between calm and stress is not statistically significant ($t = -1.46$, $p = 0.145$), so the correct conclusion is that the data provide no evidence for state-dependent relative informativeness in the direction the schedule assumes, rather than evidence for the reverse. Either way the weighting schedule has no measured basis.

The last two rows of Table 10 bear on how the negative coefficients should be read. Splitting at the holdout boundary, the fused coefficient is -0.039 ($t = -2.73$) over 2008–2023 and $+0.009$ ($t = 0.23$) over 2024–2026. The negative sign does not persist. This matters because a stable negative information coefficient would be an exploitable signal in inverted form, and the framework could be salvaged by reversing it. The holdout rules that out: the in-sample negativity is a property of that era, not a durable cross-sectional reversal effect, and inverting the composites would have bought nothing after 2023. The honest reading is that these composites carry no reliable directional information in either direction.

Two qualifications are owed. The stress regime contains only 147 sessions, so its coefficients are imprecisely estimated and the positive technical reading should not be over-interpreted. And the full-sample negative coefficients are consistent with short-horizon cross-sectional reversal in a universe of broad index funds, which is a known effect and not

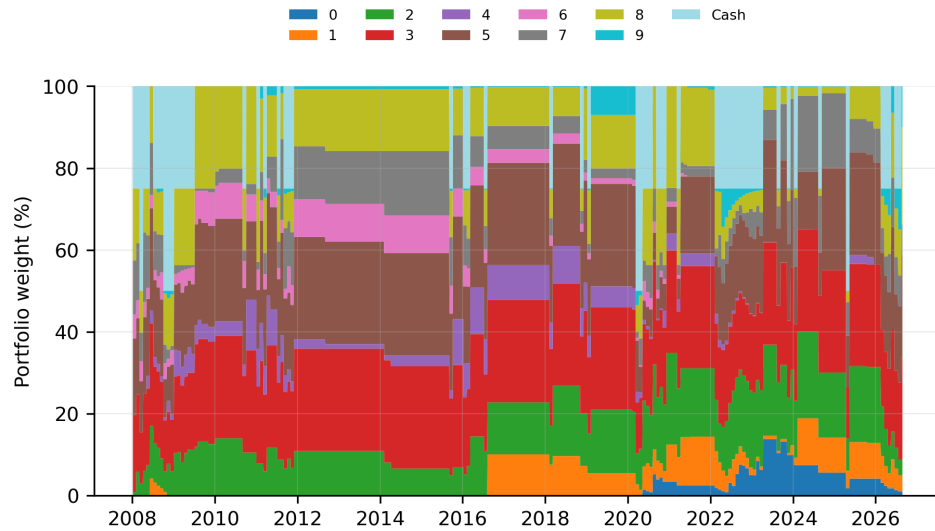


Figure 5 Portfolio weights over time, including the cash residual

evidence of an error in construction, though the holdout shows it is not stable enough to trade. The conclusion is bounded accordingly: these composites, as specified, do not carry usable directional information in this universe at this horizon.

4.9 Allocation Behaviour, Turnover and Costs

Table 11 summarises the portfolios the framework actually holds, and Figures 5 and 6 display them.

Table 11 Allocation characteristics and sensitivity to the assumed transaction cost. Effective number of assets is the reciprocal of the Herfindahl index of sleeve weights. Annual turnover is one-way, as a multiple of portfolio value.

<i>Allocation characteristics, primary specification</i>				
Mean largest weight				0.250
Mean effective number of assets				5.40
Minimum effective number of assets				4.60
Mean invested exposure				92.1%
Minimum invested exposure				50.0%
Annual one-way turnover				0.96×
<i>Sensitivity to assumed cost</i>				
Assumed cost	CAGR	Sharpe	Max DD	Turnover
0 basis points	4.90%	0.773	−19.83%	4.04×
5 basis points	5.66%	0.893	−19.54%	1.14×
10 basis points (primary)	5.65%	0.878	−18.98%	0.96×
25 basis points	5.47%	0.846	−19.30%	0.93×

The allocation is diversified and stable. The largest position sits at the 25 per cent cap on average, the effective number of assets averages 5.40 and never falls below 4.60, and annual

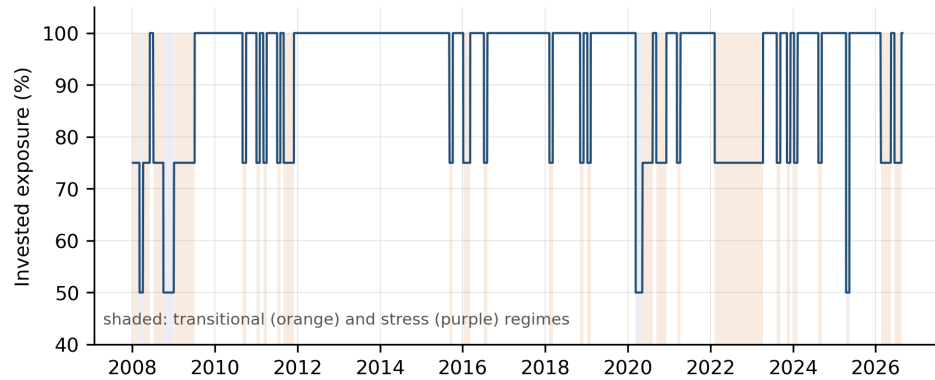


Figure 6 Invested exposure over time, with transitional and stress regimes shaded

one-way turnover is 0.96 times portfolio value, equivalent to approximately 8.0 per cent of the portfolio traded at each monthly rebalance. Invested exposure follows the regime, falling to a minimum of 50 per cent during the stress state, which is the mechanism Figure 6 displays directly.

The cost sensitivity contains a result worth stating explicitly. Performance is not monotone in the assumed cost: at zero assumed cost the Sharpe ratio is 0.773, below the 0.878 obtained at 10 basis points, even though no costs are deducted. The reason is that the assumed cost enters the objective as a turnover penalty as well as entering the return as a deduction. With the penalty removed, annual turnover rises from 0.96 to 4.04 times portfolio value as the optimiser chases small differences in the estimated covariance matrix, and the resulting weight instability costs more than the saved commissions are worth. The turnover penalty is therefore functioning as a regulariser on the weight vector, and the framework is robust across the plausible cost range of 5 to 25 basis points, over which the Sharpe ratio varies only between 0.846 and 0.893.

This section also resolves a defect in the framework as originally specified. In that version the fused score entered the objective without the information-ratio scaling of Equation (4), so the linear term was expressed in the raw units of the underlying indicators. On-balance volume takes values of order 10^8 to 10^9 , while annualised variances are of order 10^{-2} . The linear term consequently dominated the quadratic risk term by roughly ten orders of magnitude, the risk model became irrelevant to the solution, and the optimiser returned a corner solution holding a single asset on every rebalance date—an effective number of assets of exactly 1.00, with one holding persisting for 1,364 of 1,421 sessions. The scaling in Equation (4) removes the defect by placing the linear term on a return scale, and Table 11 reports the corrected behaviour. The failure was one of units, not of the optimiser or of the risk model.

4.10 Regime Model Diagnostics

Table 12 reports the mixture model's order selection statistics and the properties of the three-state solution.

The order selection statistics are reported rather than used. Both AIC and BIC continue to fall over the range examined, from 21944 and 22016 at two components to 20372 and 20599 at six, so neither criterion selects an interior optimum and neither supports the choice

Table 12 Regime model diagnostics. Order selection is computed on the full feature history; state characteristics and the transition matrix come from the three-component solution. Out-of-sample shares are the states actually assigned by the expanding-window refits used in the backtest.

<i>Order selection</i>					
Components	2	3	4	5	6
AIC	21944	20785	20571	20478	20372
BIC	22016	20896	20720	20667	20599
<i>Three-state solution</i>					
	Calm	Transitional	Stress		
Annualised volatility	12.22%	20.04%	47.07%		
Annualised momentum	+16.04%	−1.25%	−15.97%		
Share of sessions, in sample	63.6%	31.2%	5.2%		
Share of sessions, out of sample	71.4%	25.5%	3.1%		
Mean run length (sessions)	25.3	11.1	17.1		
Self-transition probability	0.961	0.910	0.941		
<i>Realised performance within each out-of-sample state</i>					
	Calm	Transitional	Stress		
Sessions	3346	1196	147		
Strategy Sharpe	1.144	0.196	1.039		
Equal-weight Sharpe	0.916	0.561	0.356		

of three components. This behaviour is expected: a mixture fitted to persistent, heavy-tailed financial features will absorb tail observations into additional components indefinitely, and the criteria reward it for doing so. The choice of three is made instead on persistence and interpretability, and Table 12 supplies the grounds. The three states are economically distinct—annualised volatility of 12.2, 20.0 and 47.1 per cent, with annualised momentum of +16.0, −1.3 and −16.0 per cent—and they persist, with self-transition probabilities between 0.910 and 0.961 and mean run lengths of 11 to 25 sessions. States of that duration can be acted upon at a monthly rebalancing frequency. Finer partitions produce shorter-lived states that the rebalancing schedule cannot exploit, and the transition matrix is close to tridiagonal—the estimated probability of moving directly between the calm and the stress state, in either direction, is zero—which indicates that the ordering recovered by sorting components on volatility is economically meaningful rather than arbitrary.

The final panel shows where the framework’s advantage arises. In the stress state the framework attains a Sharpe ratio of 1.039 against 0.356 for the benchmark, and in the calm state 1.144 against 0.916. It underperforms materially only in the transitional state, 0.196 against 0.561, which is consistent with the 2021 result in Table 7: a state that is genuinely ambiguous causes the risk budget to reduce exposure in advances that turn out to continue. Figure 7 shows the identified states over time and Figure 8 the behaviour of the two composites.

4.11 Statistical Significance

Table 13 reports inference on the daily excess return series and on the Sharpe difference.

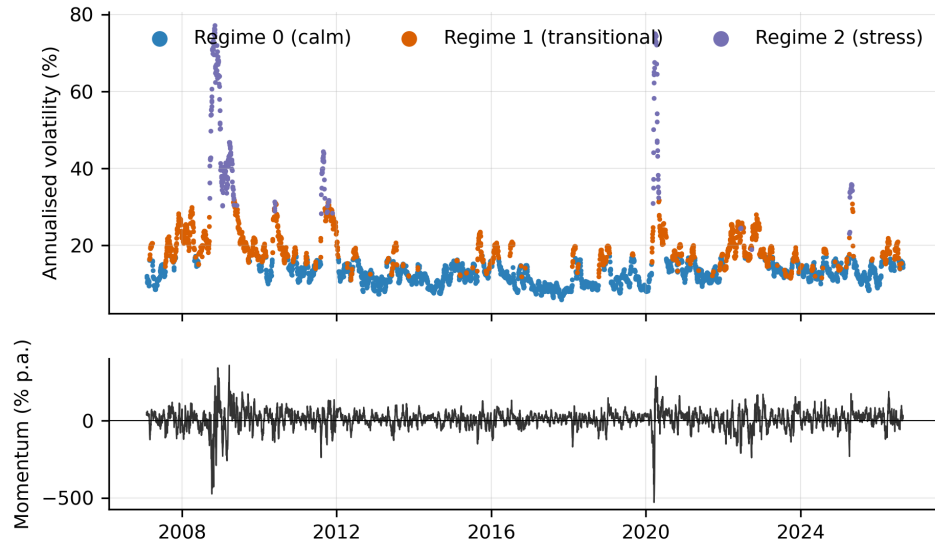


Figure 7 Identified market states, with the underlying volatility and momentum features

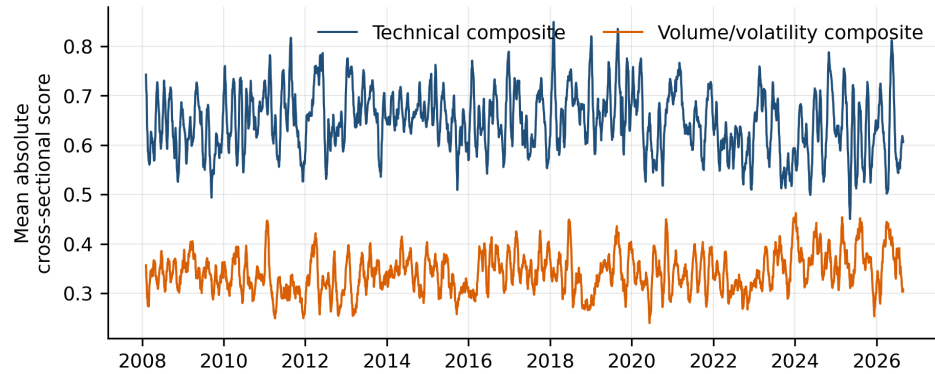


Figure 8 Mean absolute cross-sectional score of each composite, 21-session moving average

The results must be stated carefully. The framework's mean return is below the benchmark's, and that shortfall is not statistically significant ($p = 0.171$). The Sharpe difference of $+0.250$ has a bootstrap median of $+0.242$ and a 95 per cent interval of $[-0.102, 0.574]$ that includes zero; the bootstrap probability that the difference is not positive is 0.088. On an eighteen-year sample, therefore, the Sharpe improvement is not significant at conventional levels, and the paper does not claim that it is. Extending the sample by the 665 sessions of the holdout narrowed that interval—its lower bound moves from -0.194 on the pre-2024 data to -0.102 —without crossing zero, which is the expected behaviour of a real but modest effect measured on a sample that remains too short.

What is robust is not the point estimate but its consistency. The Sharpe advantage appears in 18 of 18 specifications, the drawdown advantage in 18 of 18 and in 18 of 19 calendar years, and the outperformance in all four stress episodes. Eighteen years of daily

Table 13 Inference against the equal-weighted benchmark. Bootstrap intervals are from a stationary block bootstrap with 21-session blocks and 2,000 replications.

Mean excess return over benchmark, annualised	−3.31%
Paired t -statistic	−1.369
p -value	0.171
Newey–West t -statistic, ten lags	−1.699
Strategy Sharpe ratio, 95% bootstrap interval	[0.394, 1.368]
Benchmark Sharpe ratio, 95% bootstrap interval	[0.180, 1.165]
Sharpe difference, median	+0.242
Sharpe difference, 95% bootstrap interval	[−0.102, 0.574]
Bootstrap probability that the difference is not positive	0.088

data contain few independent observations of the events that matter for a drawdown claim—four, on the evidence of Table 6—and no amount of daily sampling converts that into statistical power. The appropriate conclusion is that the drawdown reduction is consistent and mechanically explicable, and that the Sharpe improvement, while positive under every specification examined, is not established at conventional significance.

4.12 Secondary Universe: Mega-Capitalisation Equities

The framework was also run without modification on the ten mega-capitalisation US equities of the original study, over the same 2018–2026 download window, giving 1,920 evaluated sessions from January 2019. Table 14 reports the outcome, which is unfavourable on risk-adjusted return.

Table 14 Secondary universe: ten mega-capitalisation US equities, 2019–2026

	CAGR	Volatility	Sharpe	Max DD
Regime-conditional strategy	17.02%	14.87%	1.145	−20.45%
Equal-weight benchmark	33.52%	24.45%	1.371	−33.63%
Equal-weight at strategy volatility	20.08%	14.87%	1.350	−21.41%
No regime timing, same average exposure	18.82%	16.82%	1.119	−30.36%

The framework loses on risk-adjusted return, 1.145 against 1.371, and gives up 15.09 percentage points of annual return, a shortfall that is statistically significant ($t = -2.791$, $p = 0.005$). Its defensive behaviour nevertheless survives intact: maximum drawdown improves from −33.63 to −20.45 per cent, it wins on drawdown in 8 of 8 calendar years while winning on Sharpe ratio in only 3, and it outperforms in both stress episodes this window contains, by 12.50 percentage points during the 2020 crash and 12.60 percentage points during the 2022 bear market.

The regime layer itself continues to behave as the primary results describe. Against the matched-exposure control it improves the Sharpe ratio slightly, 1.145 against 1.119, and improves maximum drawdown substantially, from −30.36 to −20.45 per cent. The risk-timing effect identified in Section 4.6 therefore reproduces in a universe on which the framework loses overall, which is the relevant test of whether that effect is specific to the primary sample. What fails here is not the regime layer but the decision to hold the resulting portfolio in place of the benchmark.

The reason is a property of the universe rather than of the framework. Ten mega-capitalisation US equities are close to a single factor bet: correlations are high, there is no low-volatility asset to rotate into, and the risky sleeve therefore carries roughly 15 per cent annualised volatility against 6 per cent in the multi-asset case. De-risking in such a universe can only move capital to cash, so the framework systematically forgoes return during an exceptional advance and recovers only part of it in drawdowns. This bounds the applicability of the framework directly: it requires a universe with genuine dispersion in asset volatility, and it should not be expected to add value in a concentrated, single-sector one. The originally submitted version of this study used exactly such a universe, which is why its results are superseded here.

5 Discussion

5.1 *What the Results Establish*

The framework's contribution is narrower than the one claimed for it at the outset, and better supported. Three findings survive the evaluation of Section 4.

The first concerns the separation of the allocation decision. Splitting the problem into relative composition and total risk, rather than allowing a single constraint set to determine both, makes each part independently testable. It is this separation that allows the ablation in Table 8 to be constructed at all: without it, removing the regime layer would change both what is held and how much of it is held, and the resulting comparison would confound the two.

The second is that regime conditioning earns its place through timing, not through level. At identical average exposure, allocating risk according to the identified state rather than uniformly improves the Sharpe ratio from 0.795 to 0.878 and maximum drawdown from -22.26 to -18.98 per cent, with no significant difference in mean return ($p = 0.695$). This is the cleanest evidence in the paper, and it is available only because the control was constructed at matched risk. The corresponding cost is also measured: against a fully invested variant the framework forgoes approximately 1.3 percentage points of annual return, significantly so, in exchange for 3.6 percentage points of maximum drawdown.

The third is that the form of the risk budget matters more than its calibration. The four relative budgets in Table 9 produce Sharpe ratios within 0.020 of one another, while the absolute-target formulation—identical in intent—is inert, binding in one rebalance out of 224. An absolute volatility target is not scale-free, and a specification transferred between universes without re-calibration can silently disable the mechanism it was written to implement. This is a practical warning that generalises beyond the present framework.

5.2 *What the Results Do Not Establish*

Two claims that the framework was designed to support are not supported.

The signal fusion layer does not contribute. Its removal improves the Sharpe ratio from 0.878 to 0.908 and leaves maximum drawdown unchanged, and neither movement is significant. The information coefficients in Table 10 explain why: both composites are significantly negatively related to next-session cross-sectional returns over the full sample. The regime-conditional weighting schedule has no measured basis either. It was specified on the hypothesis that market-state information becomes relatively more informative under

stress; the point estimates run the other way and the calm-to-stress contrast is not significant ($t = -1.46$, $p = 0.145$), so the schedule is weighting on a difference the data do not establish. Splitting the sample at the holdout boundary sharpens the reading: the fused coefficient is -0.039 before 2024 and $+0.009$ after, so even the negative relationship is era-specific and could not have been exploited by inverting the composites.

The improvement in risk-adjusted return is not statistically significant. The Sharpe difference of $+0.250$ carries a bootstrap interval of $[-0.102, 0.574]$. Eighteen years of daily returns contain only a handful of independent stress events, and a claim about behaviour during market stress cannot be established at conventional significance from four episodes. Adding the 665 sessions of the holdout narrowed the interval without moving it clear of zero. The consistency of the result across 18 of 18 specifications, 18 of 19 calendar years and 4 of 4 episodes is the stronger evidence, and it is evidence of robustness rather than of significance. The two should not be conflated, and this paper does not conflate them.

5.3 Implications for Practice

For a practitioner the results suggest a specific and limited use. A regime-conditional risk budget is a reasonable component of a multi-asset allocation process where the objective includes drawdown control and where some return may be sacrificed for it. It should be implemented in relative rather than absolute form, evaluated against a volatility-matched benchmark rather than an unlevered one, and not expected to help in a universe without genuine dispersion in asset volatility.

The results also caution against a common design pattern. The framework as originally specified contained two layers whose contribution had never been isolated, one of which was inert by construction and the other of which was pointed in the wrong direction. Neither defect was visible in aggregate performance, and both were found only by ablation at matched risk. Architectures assembled from plausible components should be disassembled before they are trusted.

Finally, the turnover result in Table 11 has a practical reading. The assumed transaction cost enters the objective as a penalty as well as the return as a deduction, so setting it to zero does not produce the best achievable result—it produces a noisier weight vector and a lower Sharpe ratio. A modest cost assumption acts as a regulariser, and there is no reason to model a frictionless world even when evaluating one.

5.4 Limitations

Several limitations bound what can be concluded.

The universe is small. Ten exchange-traded funds are enough to demonstrate the mechanism but not to establish that it generalises to larger or differently composed universes, and the position cap of 25 per cent binds on average, which means the optimiser's freedom is partly determined by the constraint rather than by the risk model.

The evaluation is a single historical path. Eighteen years is a long sample by the standards of this literature and a short one for inference about tail events, and the four stress episodes are not independent draws from a stationary process.

The signals are market-derived proxies rather than textual sentiment. No news, filings or social media text is used at any point, as stated in Section 3.3. The null reported in Section 4.7 is therefore a null about these proxies, and it is not evidence that textual sentiment would fail in the same role. Establishing that would require a time-stamped news corpus with

coverage of the traded instruments, which for broad exchange-traded funds does not exist at the instrument level.

The regime model is deliberately simple. A three-component Gaussian mixture on two features is interpretable and stable, but it imposes conditional normality within states, treats observations as conditionally independent given the state, and—as Table 12 shows—is not the specification either information criterion would select.

The risk-free rate is taken as zero throughout, and cash is assumed to earn nothing. Over a sample containing both the near-zero rates of 2009–2015 and the higher rates of 2023, this understates the return on the framework’s cash holding and therefore understates its performance relative to the fully invested benchmark. The direction of this bias is against the framework, which is the conservative choice, but it is a simplification.

5.5 *Future Work*

Four extensions follow directly. The first is textual sentiment, applied to a universe where an instrument-level news corpus exists, which would convert the proxy null of Section 4.7 into a genuine test of the fusion hypothesis. The second is a larger and more heterogeneous universe, including commodities, currencies and non-US fixed income, which would test whether the volatility-dispersion condition identified in Section 4.11 is the binding requirement it appears to be. The third is learning the risk-budget multipliers rather than fixing them, for which reinforcement learning is a natural candidate given the sequential structure of the problem, subject to the overfitting concerns documented by Harvey and Liu (2015) and Valls and Steurer (2026). The fourth is replacing the mixture model with a formulation that represents state persistence explicitly, such as a hidden Markov or jump model (Li et al., 2025), which would remove the conditional-independence assumption noted above.

6 Conclusion

This paper set out to close the step between a predictive signal and an executable portfolio, and to determine which parts of a regime-conditional framework are responsible for its behaviour. Both objectives are met, though not in the direction originally anticipated.

The framework, evaluated on ten liquid multi-asset instruments over eighteen years under a walk-forward protocol with transaction costs, attains a Sharpe ratio of 0.878 against 0.628 for an equal-weighted benchmark and reduces maximum drawdown from -36.1 to -19.0 per cent, while earning a lower compound return. It outperforms in all four major stress episodes in the sample, improves on drawdown in 18 of 19 calendar years, and beats the benchmark on both measures in all 18 specifications examined. The final 665 sessions, from January 2024 to August 2026, postdate every specification decision and were held out; on them the framework records a higher Sharpe ratio and a smaller maximum drawdown than all three benchmarks. The Sharpe improvement is nevertheless consistent rather than statistically significant: its bootstrap interval includes zero.

Ablation locates the source of that behaviour precisely. At identical average exposure, timing risk by the identified market state improves both the Sharpe ratio and the maximum drawdown with no difference in mean return, which is the signature of a genuine risk-timing effect. The form of the risk budget proves more consequential than its calibration: expressed as a fraction of the portfolio’s own volatility the layer is active and effective, while the

absolute volatility targets of the original specification bind in one rebalance out of 224 and leave it inert.

The signal fusion layer, which gave the framework its name, does not contribute. Removing it improves the Sharpe ratio, and the information coefficients of both composites are significantly negative over the full sample. The regime-conditional weighting schedule has no measured basis: it was specified on the hypothesis that market-state information gains importance under stress, and the calm-to-stress contrast is not significant ($p = 0.145$) with point estimates pointing the other way. Nor is the negative relationship stable enough to invert, since it disappears in the holdout. This null is reported in full, with the coefficients that produce it, because a component-level negative result is more useful than an aggregate number that conceals which parts of an architecture are doing the work.

The study's contribution is therefore a specification and a method of testing it: a two-step separation of composition from total risk that makes each layer independently falsifiable, an ablation design that isolates risk timing from risk level by matching average exposure, and a documented account of two failure modes—an inert absolute risk budget, and a signal-to-allocation conversion in which mismatched units rather than information determine the solution. The framework is useful within stated bounds: it requires a universe with genuine dispersion in asset volatility, it trades return for drawdown at a measured rate, and it should be evaluated against a volatility-matched benchmark rather than an unlevered one. Outside those bounds, as the mega-capitalisation results show, it does not add value.

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Author contributions

Anandkumar Pardeshi: Conceptualization, Methodology, Software, Data curation, Formal analysis, Writing – original draft.

Sujata Deshmukh: Supervision, Validation, Writing – review & editing, Research coordination.

Conflict of interest

The authors declare no conflict of interest.

Data and code availability

All market data used in this study are publicly available daily price and volume series obtained from Yahoo Finance for the instruments listed in Section 3.1. The analysis code implementing the framework, the specification sweep and the ablations is available from the corresponding author on reasonable request.

AI tool usage declaration

During the preparation of this work, the authors used [AUTHORS: NAME THE TOOL(S) ACTUALLY USED] in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. No part of the analysis, the generation of results, or the interpretation of findings was delegated to such tools.

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